

Convertible Bond Pricing with Longstaff-Schwartz

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Abstract:

Convertible bonds are hybrid securities that combine features of debt and equity, making their valuation challenging due to the embedded optional conversion. This paper applies the Longstaff-Schwartz least-squares Monte Carlo (LSMC) algorithm to price a convertible bond, to provide an exposition of the method's implementation and behaviour. Monte Carlo simulation with a regression-based optimal stopping strategy offers flexibility in handling the American-style conversion option and potential path-dependent features, in contrast to lattice or finite-difference approaches which struggle with complex call or put schedules and multiple state variables.

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1 Introduction

Convertible bonds are hybrid securities that offer an option to convert the bond into a predetermined number of shares of the issuer's stock. In other words, a convertible bond is a corporate bond with a call option on the company's stock. If the company's stock performs well, the bondholder can convert into shares and benefit from the equity appreciation; if the stock performs poorly, the investor can hold the bond and receive regular interest and the face value at maturity.

Every convertible bond has certain key parameters that determine its value and conversion mechanics. The *face value* is the principal amount of the bond that will be returned to the investor at maturity if not converted. Additionally, the *conversion ratio* is the number of shares of stock that the bondholder is entitled to receive upon converting one bond. From these definitions, we define the *conversion price* as the effective price per share at which conversion happens. If we let the bond's face value be denoted as F and N is the conversion ratio, then conversion price is defined as

$$\mathcal{P} = \frac{F}{N}.$$

Another important metric is the *conversion value*, also known as *parity*, which is the current market value of the shares into which the bond can be converted. Mathematically speaking, $\text{parity} = S \times N$, where S is the current stock price. It represents the value the bondholder would get by converting the bond at the prevailing stock price. On the other hand, the bond's *straight value* is the value of the bond without the conversion option. Essentially, the present value of its promised cashflows (coupons plus principal) discounted at the yield of a comparable non-convertible bond. The straight value (also known as *bond floor*) represents a lower bound for the convertible's price, since it reflects what the bond would be worth as a pure debt instrument. Hence, a convertible's market price is composed of two components: the straight bond value and the option component. The *conversion premium* is the amount by which the convertible's market price exceeds its conversion value (parity). Because the investor can always choose not to convert and instead hold the bond to maturity, the convertible's price should never fall below the greater of its straight-bond value and its conversion value. Arbitrage arguments guarantee that a convertible bond will trade at at least the higher of its bond floor and its parity value. If it were to trade below that level, an arbitrage opportunity exists, and one could buy the underpriced convertible and either hold it as a bond or convert it to stock to realise an instant profit.¹ Hence, the minimum value of a convertible at any time is $\max(\text{straight bond value}, \text{conversion value})$.

Since convertibles consist of a stock option, their value and behaviour depend on the *moneyness* of the conversion option. The conversion option is considered *in-the-money* (ITM) if the current stock price exceeds the conversion price, i.e., $S > \mathcal{P} \iff SN > F$. Conversely, it is *out-of-the-money* (OTM) if the stock price is below the conversion price, $S < \mathcal{P}$, making immediate conversion irrational. If $S \approx \mathcal{P}$, then the option is near *at-the-money* (ATM). When the option is deeply ITM ($S \gg \mathcal{P}$), the convertible bond's price is largely driven by its equity value, and it behaves like an equity-like instrument. Here, the bond's market price will be close to the conversion value and will move nearly one-for-one with the underlying stock, with only a small premium reflecting any remaining

¹Mathematical proof in Appendix 1

bond interest or optionality. In the opposite case, when the option is far OTM ($S \ll \mathcal{P}$), the convertible behaves like a bond-like instrument. Here, the stock price is so low that conversion is unlikely, so the convertible's value moves toward its bond floor and is more sensitive to interest rates and credit risk than to the stock price. Finally, the case at ATM is often referred to as a *balanced convertible*, where the stock price is in around the conversion price ($S \approx \mathcal{P}$) such that the convertible has significant bond value and significant option value at the same time. In this region, the convertible's price will consist of bond characteristics as well as equity option upside, meaning it reacts to both the underlying movements and interest rate/credit changes.

In extreme situations where the issuer's stock price has fallen very sharply and the company's credit has dropped, a convertible can trade at *distressed* levels. Distressed convertible is a bond whose conversion option is practically worthless (the stock is deeply OTM), such that the issue trades almost purely on its remaining bond value. These convertibles often exhibit high yields, due to the issuer's elevated credit risk, and a significantly high conversion premium, since the market price of the bond significantly exceeds the negligible equity conversion value. Hence, a distressed convertible is one that has become debt-like in value due to a collapsed stock price and investor concerns about the issuer's solvency. Under these circumstances, the bond may trade at a substantial discount, but it also means that if the company improves and the stock rebounds, the convertible has a lot of upside potential from those low levels.

Due to the nature of convertibles, it is rare to find European-style convertible bonds. Realistically, convertible bonds are structured with American-style (or Bermudan-style) derivative, whereby the user can convert at any point between the issue of the bond and the maturity date. American-style derivatives present more challenge than a European-style derivative to price, since one cannot take the fair value of the derivative at maturity. Instead, at each point during the lifetime of the derivative, one must compare the value of immediate exercise with the expected value of holding the derivative, with the opportunity to exercise later. Traditional valuation approaches for convertibles proceed either by discrete-time lattices. In a binomial or trinomial lattice the stock process is discretised, and at each node one solves a backward induction with an early-exercise comparison. This method as well as other such as finite-difference methods encounter practical issues for realistic contracts. For example, callable schedules, soft-call provisions, coupon accrual conventions, and defaultable cashflow legs induce path dependence. Stochastic interest rates and stochastic credit spreads introduce additional state variables, and equity dynamics may include jumps or local stochastic volatility. Consequently, classical grid methods can become impractical in American-style optionality coupled with multiple stochastic drivers and event-driven payoffs.

Monte Carlo simulation can help with dimensionality and accommodates path dependence, however a naïve simulation of discount payoffs does not resolve the optimal stopping inherent in conversion. Formally, if we let $(\mathcal{F}_t)_{t \in [0, T]}$ denote the market filtration and $\mathbb{Q}$ the risk-neutral measure, then the convertible's value at time t solves the Snell envelope representation

$$V_t = \operatorname{ess\,sup}_{\tau \in \mathcal{T}_{t, T}} \mathbb{E}_{\mathbb{Q}} \left[e^{-\int_t^\tau r_u du} X_\tau \mid \mathcal{F}_t \right],$$

where $\mathcal{T}_{t, T}$ is the set of $\mathcal{F}_t$ stopping times taking values in $[t, T]$, r is the short rate, and X_τ is the cashflow at the first event (conversion, call, default, or maturity) occurring at τ . A

forward simulation samples $\{X_\tau\}$ given a fixed policy τ , and as such, it does not learn the maximising policy. What is missing is a way to estimate, from cross-sectional simulation data, the conditional expectation of continuation that defines the optimal policy through the dynamic programming principle.

The *Longstaff-Schwartz least-squares Monte Carlo* (LSM) method addresses this by using least-squares to estimate the conditional expected value of continuing to hold the derivative at each time step. On a discrete grid $0 = t_0 < t_1 < \dots < t_n = T$, where n denotes number of time steps, define the continuation value at time t_i by

$$C(t_i, S_{t_i}) = \mathbb{E}_{\mathbb{Q}} \left[e^{-\int_{t_i}^{t_{i+1}} r_u du} V_{t_{i+1}} \mid S_{t_i} \right],$$

where S is the equity state.

In this paper, we use formalise conditional expectation as an L^2 projection and derive the associated least-squares estimators before presenting the LSM algorithm and applying it to a non-callable, non-puttable, zero-coupon convertible bond under geometric Brownian motion. We then document the numerical results and interpret the estimated continuation values, stopping behaviour, and conversion boundary generated by the LSM procedure.

2 Conditional Expectations and Least-Squares Regression

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For a sub-sigma-algebra $\mathcal{G} \subseteq \mathcal{F}$, if $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$, then the conditional expectation $\mathbb{E}[X \mid \mathcal{G}]$ is the unique $\mathcal{G}$ -measurable random variable in L^2 that minimises the mean squared error among all $\mathcal{G}$ -measurable square-integrable random variables Y . In other words,

$$\mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])^2] \leq \mathbb{E}[(X - Y)^2], \quad Y \in L^2, Y \text{ that is } \mathcal{G}\text{-measurable.}$$

Equivalently, $\mathbb{E}[X \mid \mathcal{G}]$ is the orthogonal projection of X onto the closed subspace of $\mathcal{G}$ -measurable square integral random variables, and therefore satisfies the condition: $\mathbb{E}[(X - \mathbb{E}[X \mid \mathcal{G}])Z] = 0$ for all $\mathcal{G}$ -measurable $Z \in L^2$.

The link between conditional expectation and least-squares regression is fundamental to the LSM method. Consider two square-integrable random variables X and Y . The conditional expectation $\mathbb{E}[Y \mid X]$ is a function of X (measurable w.r.t. X) which can be thought of as the best predictor of Y given X . In fact, we can prove that $\mathbb{E}[Y \mid X]$ is the unique function of X that achieves the minimum mean squared error $\mathbb{E}[(Y - f(X))^2]$ among all measurable functions f . In other words,

$$\mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2] \leq \mathbb{E}[(Y - f(X))^2], \quad f \text{ measurable with } \mathbb{E}[f(X)^2] < \infty. \quad (1)$$

We can prove this by looking at the right hand side of (1):

$$\begin{aligned} \mathbb{E}[(Y - f(X))^2] &= \mathbb{E}[(Y - f(X) + \mathbb{E}[Y \mid X] - \mathbb{E}[Y \mid X])^2] \\ &= \mathbb{E}[((Y - \mathbb{E}[Y \mid X]) - (f(X) - \mathbb{E}[Y \mid X]))^2] \\ &= \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2 + (f(X) - \mathbb{E}[Y \mid X])^2 - 2(Y - \mathbb{E}[Y \mid X])(f(X) - \mathbb{E}[Y \mid X])], \end{aligned}$$

and by linearity of expectations,

$$= \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2] + \mathbb{E}[(f(X) - \mathbb{E}[Y \mid X])^2] - 2 \mathbb{E}[(Y - \mathbb{E}[Y \mid X])(f(X) - \mathbb{E}[Y \mid X])].$$

If we look at the final term, we can apply the tower property to get,

$$\begin{aligned}\mathbb{E}[(Y - \mathbb{E}[Y | X])(f(X) - \mathbb{E}[Y | X])] &= \mathbb{E}[\mathbb{E}[(Y - \mathbb{E}[Y | X])(f(X) - \mathbb{E}[Y | X]) | X]] \\ &= \mathbb{E}[\mathbb{E}[Y f(X) - Y \mathbb{E}[Y | X] - f(X) \mathbb{E}[Y | X] + (\mathbb{E}[Y | X])^2 | X]]\end{aligned}$$

Using the property that $\mathbb{E}[g(X)Y | X] = g(X) \mathbb{E}[Y | X]$:

$$\begin{aligned}&= \mathbb{E}[f(X) \mathbb{E}[Y | X] - (\mathbb{E}[Y | X])^2 - f(X) \mathbb{E}[Y | X] + (\mathbb{E}[Y | X])^2] \\ &= \mathbb{E}[f(X) \mathbb{E}[Y | X] - f(X) \mathbb{E}[Y | X]] = 0.\end{aligned}$$

Hence,

$$\mathbb{E}[(Y - f(X))^2] = \mathbb{E}[(Y - \mathbb{E}[Y | X])^2] + \mathbb{E}[(f(X) - \mathbb{E}[Y | X])^2] - 2 \times 0.$$

Since, $\mathbb{E}[(f(X) - \mathbb{E}[Y | X])^2] \geq 0$,

$$\mathbb{E}[(Y - f(X))^2] \geq \mathbb{E}[(Y - \mathbb{E}[Y | X])^2].$$

In other words,

$$\mathbb{E}[Y | X] = \arg \min_{f(X)} \mathbb{E}[(Y - f(X))^2]. \quad (2)$$

In practice, one approximates this optimal function by restricting to a class of functions (such as polynomials in X or other basis functions) and performing a least-squares regression on historical data. Suppose we choose finite set of basis functions $\{\phi_1(x), \phi_2(x), \dots, \phi_m(x)\}$, for example powers of x , $\{1, x, x^2, \dots, x^m\}$ to span an approximation space. We then approximate conditional expectation of the form

$$f(X) = \alpha_1 \phi_1(X) + \alpha_2 \phi_2(X) + \dots + \alpha_m \phi_m(X) = \sum_{j=1}^m \alpha_j \phi_j(X),$$

that best fits Y in the least-squares sense (Longstaff et al., 2001). The regression problem is to find coefficients $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_m)$ minimising

$$R(\boldsymbol{\alpha}) = \mathbb{E} \left[\left(Y - \sum_{j=1}^m \alpha_j \phi_j(X) \right)^2 \right].$$

We can write the residual as $\epsilon(\boldsymbol{\alpha}) = Y - \boldsymbol{\alpha}^\top \boldsymbol{\phi}(X)$, then

$$\begin{aligned}R(\boldsymbol{\alpha}) &= \mathbb{E}[\epsilon(\boldsymbol{\alpha})^2] = \mathbb{E}[(Y - \boldsymbol{\alpha}^\top \boldsymbol{\phi}(X))^2] \\ &= \mathbb{E}[Y^2 - 2Y \boldsymbol{\alpha}^\top \boldsymbol{\phi}(X) + (\boldsymbol{\alpha}^\top \boldsymbol{\phi}(X))^2],\end{aligned}$$

by linearity

$$R(\boldsymbol{\alpha}) = \mathbb{E}[Y^2] - 2\mathbb{E}[Y(\boldsymbol{\alpha}^\top \boldsymbol{\phi}(X))] + \mathbb{E}[(\boldsymbol{\alpha}^\top \boldsymbol{\phi}(X))^2].$$

By using the fact that $(A^\top B)(B^\top A) = A^\top (BB^\top)A$ and linearity,

$$R(\boldsymbol{\alpha}) = \mathbb{E}[Y^2] - 2\boldsymbol{\alpha}^\top \mathbb{E}[\boldsymbol{\phi}(X)Y] + \boldsymbol{\alpha}^\top \mathbb{E}[\boldsymbol{\phi}(X)\boldsymbol{\phi}(X)^\top] \boldsymbol{\alpha}.$$

Define $\varphi := \mathbb{E}[\boldsymbol{\phi}\boldsymbol{\phi}^\top]$, $b := \mathbb{E}[\boldsymbol{\phi}(X)Y]$, and $c := \mathbb{E}[Y^2]$, then the risk is the quadratic form $R(\boldsymbol{\alpha}) = c - 2\boldsymbol{\alpha}^\top b + \boldsymbol{\alpha}^\top \varphi \boldsymbol{\alpha}$. We can note that c, φ, b do not depend on $\boldsymbol{\alpha}$ and b, φ depend

on the joint law of (X, Y) and the choice of basis. Differentiating $R(\boldsymbol{\alpha})$ with respect to $\boldsymbol{\alpha}$ gives us

$$\nabla_{\boldsymbol{\alpha}} R(\boldsymbol{\alpha}) = -2b + 2\varphi\boldsymbol{\alpha}.$$

Setting this to zero gives us $\varphi\boldsymbol{\alpha} = b$. This linear system is the population normal equations. If the chosen basis functions are linearly independent in L^2 , meaning that $\beta^\top \phi = 0$ a.s. $\implies \beta = 0$, then φ is positive definite and therefore invertible, hence

$$\boldsymbol{\alpha} = \varphi^{-1}b.$$

In the context of the Longstaff-Schwartz method, Y will represent the realised future payoff from continuation of an option, and X will be the state variable(s) at the current time (such as the underlying stock price). By regressing Y on functions of X across many simulated paths, we can obtain an estimate of the continuation value, $\mathbb{E}[Y | X]$, without needing an analytic formula.

3 Least-Squares Monte Carlo (LSM) for Convertible Bond Pricing

Convertible bonds in practice can be converted into the predetermined shares at any point of its lifetime, hence acting as an American-style derivative. Due to the complexity of modelling continuous exercise time, we will model convertible bonds under a *Bermudan-style derivative*, which allows us to exercise the conversion over a fixed and finite number of times up to and including maturity. We will split this section into two parts: the first is an introduction to LSM for a Bermudan option. The second part will be about LSM for convertible bonds.

3.1 Bermudan Option

Say we have a Bermudan option V with $n \in \mathbb{N}$ exercise times $0 < t_1 < t_2 < \dots < t_n = t$, then define the immediate exercise value for a Bermudan at t_i as

$$h_i(S_{t_i}) := (S_{t_i} - K_i)^+,$$

where K_i is the strike at time t_i and S is the underlying. Then the option value at time t_i satisfies the dynamic programming recursion

$$V_{t_i} = \max \left(h_i(S_{t_i}), e^{-r(t_{i+1}-t_i)} \mathbb{E}_{\mathbb{Q}}[V_{t_{i+1}} | \mathcal{F}_{t_i}] \right), \quad i = 1, \dots, n-1. \quad (3)$$

Thus, at each exercise date, the holder compares the immediate exercise payoff with the continuation value. If immediate exercise is optimal, the option is exercised; otherwise it is continued to the next exercise date. As the exercise mesh becomes dense, i.e., $\max_i(t_{i+1} - t_i) \rightarrow 0$, the Bermudan problem. For the paths where the derivative is ITM, we only exercise if and only if the exercise is more profitable than continuation. Hence, to calculate the fair price at time t_i ($i \neq n$) of a Bermudan-style derivative, we take the maximum value of either exercising at time t_i or hold to exercise at t_{i+1} . Paths that are ATM or OTM are automatically not profitable to exercise at time t_i and the investor will choose to hold the derivative instead. Hence the value of ATM and OTM the derivative at time t_i will be the same value as the derivative at time t_{i+1} discounted back to t_i .

To calculate the conditional value $e^{-r(t_{i+1}-t_i)} \mathbb{E}_{\mathbb{Q}}[V_{t_{i+1}} \mid \mathcal{F}_{t_i}]$ is generally not available in closed form, because the holder retains multiple future exercise opportunities. Therefore, unlike the European case, there is no simple analytic pricing formula such as Black-Scholes that directly gives the discounted continuation value.

One possible method that was designed to tackle this was through nested or resimulation-based Monte Carlo. In that method, we launch a new conditional Monte Carlo simulation from time t_i along each path in order to estimate the continuation value. However, this quickly becomes computationally impractical, since the number of required simulations grows very rapidly with the number of exercise dates, in fact, the required number of paths grows exponentially by $(n_{\text{paths}})^{n_{\text{exercise}}}$.

A solution to this could be to use *binning*. For example, say we had a Bermudan option with two exercise dates $t_1 < t_2$. At time t_1 , we want to compare the immediate exercise payoff $(S_{t_1} - K_1)^+$ with the continuation value $e^{-r(t_2-t_1)} \mathbb{E}_{\mathbb{Q}}[(S_{t_2} - K_2)^+ \mid \mathcal{F}_{t_1}]$, where K_1, K_2 are the strikes at the two exercise dates. The optimal exercise in t_1 compares the exercise value with the value of the t_2 option:

$$e^{-r(t_2-t_1)} \mathbb{E}[V_{t_2} \mid \mathcal{F}_{t_1}].$$

If we take the risk-neutral stock price to be modelled under a geometric Brownian motion:

$$dS_t = rS_t dt + \sigma S_t dW_t,$$

it is then sufficient to calculate

$$e^{-r(t_2-t_1)} \mathbb{E}_{\mathbb{Q}}[(S_{t_2} - K_2)^+ \mid S_{t_1}].$$

We can then use binning to calculate an approximation to the conditional expectation. For a given S_{t_1} none or very few values of the continuation values exist. An estimate is not possible or else exhibits a *foresight* bias, i.e, non-measurable bias. For an interval $[S_1 - \epsilon, S_1 + \epsilon]$ with sufficiently large ϵ we have enough values to estimate

$$e^{-r(t_2-t_1)} \mathbb{E}_{\mathbb{Q}}[(S_{t_2} - K_2)^+ \mid S_{t_1} \in [S_1 - \epsilon, S_1 + \epsilon]],$$

which in turn may be used as an estimate for

$$e^{-r(t_2-t_1)} \mathbb{E}_{\mathbb{Q}}[(S_{t_2} - K_2)^+ \mid S_{t_1} = S_1].$$

Now consider the binning as a piecewise-constant approximation of the conditional expectation $e^{-r(t_2-t_1)} \mathbb{E}_{\mathbb{Q}}[V_{t_2} \mid Z]$. Let $\{U_i\}_i$ be a collection of disjoint bins partitioning the state space of Z . On each bin U_i , define the constant

$$H := e^{-r(t_2-t_1)} \mathbb{E}_{\mathbb{Q}}[V_{t_2} \mid Z \in U_i].$$

The the piecewise-constant estimator $H(Z) = \sum_i H_i \mathbf{1}_{\{Z \in U_i\}}$ is the least-squares projection of the discounted future payoff onto the class of functions that are constant on each bin. Equivalently, if $\mathcal{H}$ denotes the set of all functions of the form

$$G(Z) = \sum_i g_i \mathbf{1}_{\{Z \in U_i\}},$$

then H solves

$$H = \arg \min_{G \in \mathcal{H}} \mathbb{E}_{\mathbb{Q}} \left[\left(e^{-r(t_2-t_1)} V_{t_2} - G(Z) \right)^2 \right]. \quad (4)$$

The problem about binning stems from the partitioning of the state space $Z(\Omega)$ into finite number of bins. This results in a piecewise constant approximation of the conditional expectation. We can solve this issue by approximating the conditional expectation using a smooth function of the state variable Z . We can do this via a *least-squares approximation* of the conditional expectation.

To approximate the continuation value, Longstaff-Schwartz projects it onto a finite-dimensional span of basis functions of the current state variables. Let $Y = (Y_1, \dots, Y_p)$ be a vector of $\mathcal{F}_{t_1}$ -measurable state variables, and let

$$f(Y, \alpha) = \sum_{j=1}^q \alpha_j \phi_j(Y),$$

where $\{\phi_j\}_{j=1}^q$ is a chosen family of basis functions. Given simulated paths $\Omega^* = \{\omega_1, \dots, \omega_m\}$ and target continuation values $U(\omega_1), \dots, U(\omega_m)$, the least-squares estimator chooses α^* to minimise

$$\sum_{\ell=1}^m (U(\omega_\ell) - f(Y(\omega_\ell), \alpha))^2.$$

In matrix form if

$$X = \begin{pmatrix} \phi_1(Y(\omega_1)) & \cdots & \phi_q(Y(\omega_1)) \\ \vdots & \ddots & \vdots \\ \phi_1(Y(\omega_m)) & \cdots & \phi_q(Y(\omega_m)) \end{pmatrix}, \quad u = \begin{pmatrix} U(\omega_1) \\ \vdots \\ U(\omega_m) \end{pmatrix},$$

then the normal equations are

$$X^\top X \alpha^* = X^\top u.$$

If $X^\top X$ is invertible, the least-squares solution is

$$\alpha^* = (X^\top X)^{-1} X^\top u.$$

This fitted function is then used as an approximation to the continuation value at time t_i .

3.2 Convertible Bond

Unlike Bermudan options, convertible bonds have interim cashflows (coupons), which means that we must change the discounted value from $V_{T_{i+1}}$ to $V_{T_{i+1}} + c_{i+1}$, for some coupon c_{i+1} . When considering the exercise we take the conversion at time t_i to be $X_i^{\text{conv}} = NS_{t_i}$, giving us

$$V_{t_i} = \max(NS_{t_i}, C(t_i, S_{t_i})), \quad C(t_i, S_{t_i}) = \mathbb{E}_{\mathbb{Q}}[Y_{i+1} \mid S_{t_i}], \quad Y_{i+1} = e^{-r\Delta t_i}(V_{t_{i+1}} + c_{i+1}).$$

We note that this specification assumes the underlying state variable is Markov and observable at time t_i , so that the continuation value can be expressed as a function of the current stock price. We estimate

$$C(t_i, S_{t_i}) = \mathbb{E}_{\mathbb{Q}}[Y_{i+1} \mid S_{t_i}],$$

by regressing the realised discounted continuation cashflows $Y_{i+1}^{(k)}$ on a Laguerre basis evaluated at the current state variable $S_{t_i}^{(k)}$. In this way, the regression provides a least-squares estimate to the conditional expectation and hence to the continuation value

$C(t_i, S_{t_i})$. Like with the Bermudan option, we want to represent $C(t_i, S_{t_i})$ as a basis function. This basis function is square-integrable as we are minimising the mean squared error, which only well-defined and finite if the second moment exist. In other words, we want this conditional expectation to be an approximate L^2 projection, hence want the basis function to be of L^2 . In practice, a common option to use is the *Laguerre Polynomials*, given as

$$L_n(x) = \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{x^k}{k!}.$$

We can represent the conditional value as

$$C(t_i, S_{t_i}) \approx \sum_{j=0}^m \alpha_j L_j \left(\frac{S_{t_i}}{\mathcal{P}} \right),$$

where the coefficients α_j are to be determined by least squares regression. For some degree $m \in \mathbb{N}$ of the Laguerre polynomials that we want to approximate the conditional expectation for, we can choose $\boldsymbol{\alpha} = (\alpha_0, \dots, \alpha_m)$ that minimises the mean squared error

$$\boldsymbol{\alpha}^* = \arg \min_{\boldsymbol{\alpha} \in \mathbb{R}^{m+1}} \mathbb{E}_{\mathbb{Q}} \left[\left(Y_{i+1} - \sum_{j=0}^m \alpha_j L_j \left(\frac{S_{t_i}}{\mathcal{P}} \right) \right)^2 \right].$$

Under Monte Carlo simulation with simulated paths $k = 1, \dots, M$, this population is represented by the sample least-squares problem

$$\hat{\boldsymbol{\alpha}}^* = \arg \min_{\boldsymbol{\alpha} \in \mathbb{R}^{m+1}} \sum_{k=1}^M \left(Y_{i+1}^{(k)} - \sum_{j=0}^m \alpha_j L_j \left(\frac{S_{t_i}^{(k)}}{\mathcal{P}} \right) \right)^2, \quad (5)$$

so that $\hat{C}(t_i, S_{t_i})$ becomes the fitted value from regressing the discounted future cashflows $Y_{i+1}^{(k)}$ on the Laguerre basis evaluated at $S_{t_i}^{(k)}$.

4 Results

4.1 Non-callable, Non-puttable Convertible Bond

We implement the LSM stopping rule by comparing the immediate conversion value with the estimated continuation value at each exercise date. If the bond is sufficiently ITM and immediate conversion dominates continuation, the bond is converted; otherwise it is held to the next time step. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be our probability space with the filtration $\{\mathcal{F}_t\}_{t \in [0, T]}$, where T is maturity of the convertible bond. We will model our underlying process $\{S_t\}_{t \in [0, T]}$ as a Geometric Brownian Motion with respect to the probability space, with the risk-free asset, volatility, and dividend yields as r , σ , and q respectively. Under a risk-neutral measure, we can simulate the stock price using the Euler-Maruyama discretisation

$$S_{t+\Delta t} = S_t + (r - q)S_t \Delta t + \sigma S_t \sqrt{\Delta t} Z, \quad Z \sim \mathcal{N}(0, 1).$$

Using the same variables defined in subsection 3.2, we can note that we exercise if $NS_{t_i} > C(t_i, S_{t_i}) \approx \hat{C}(t_i, S_{t_i})$, otherwise we hold to t_{i+1} , for $0 = t_0 < t_1 < \dots < t_{n-1} < t_n = T$.

We calculate the value of the convertible bond by using backwards induction. We begin at time $t_n = T$. After the last exercise time, the value of the future payments is $C(t_n, S_{t_n}) = 0$. Our induction step we set $t = t_i$, which gives us:

$$\begin{cases} V_{t_i} = NS_{t_i}^{(k)}, & \text{if } NS_{t_i}^{(k)} \geq \hat{C}(t_i, S_{t_i}^{(k)}), \\ Y_{i+1}, & \text{otherwise.} \end{cases}$$

where $Y_{i+1} = e^{-r\Delta t_i}(V_{t_{i+1}}^{(k)} + c_{i+1})$, N is the number of predetermined shares upon exercise, and $c_{i+1} = 0$.² It is important to note that $\hat{C}(t_i, S_{t_i}^{(k)}) \approx \text{Regress}\left(Y_{i+1}^{(k)} \text{ on basis}(S_{t_i}^{(k)})\right)$. Additionally, we calculate the value of $C(t_i, S_{t_i})$ through the regression problem given in (5), with the scaled regressor $S_{t_i}^{(k)}/\mathcal{P}$.

Finally, at time t_0 the convertible price is estimated by the Monte Carlo average

$$V_0 \approx \frac{1}{M} \sum_{k=1}^M V_{t_0}^{(k)}.$$

In the experiment, we consider a zero-coupon, non-callable, non-puttable convertible bond with face value $F = 100$ and conversion ratio $N = 1$. Hence the conversion price is $\mathcal{P} = F/N = 100$. The stock is initially set to $S_0 = 100$, so that the contract is at-the-money at inception, with parameters $(r, q, \sigma) = (0.04, 0, 0.1)$ and maturity $T = 2$ years. The LSM continuation is approximated using a Laguerre basis scaled by $S/\mathcal{P}$. Since $q = 0$ here, early conversion is not economically attractive through foregone cash-flows. converting early replaces the convex claim on the stock, that being the embedded option-like component, with the linear payoff NS_t , and therefore sacrifices time value. In continuous-time setting, this is analogous to the familiar *no early exercise* property of a non-dividend-paying American call. As a result, we expect most paths to delay conversion until maturity, and the price to be close to the European decomposition

$$V_0 = \mathbb{E}_{\mathbb{Q}} [e^{-rT} \max(NS_T, F)] = Fe^{-rT} + \mathbb{E}_{\mathbb{Q}} [e^{-rT} (NS_T - F)^+].$$

²zero coupon bond $c_{i+1} = 0, \forall i$

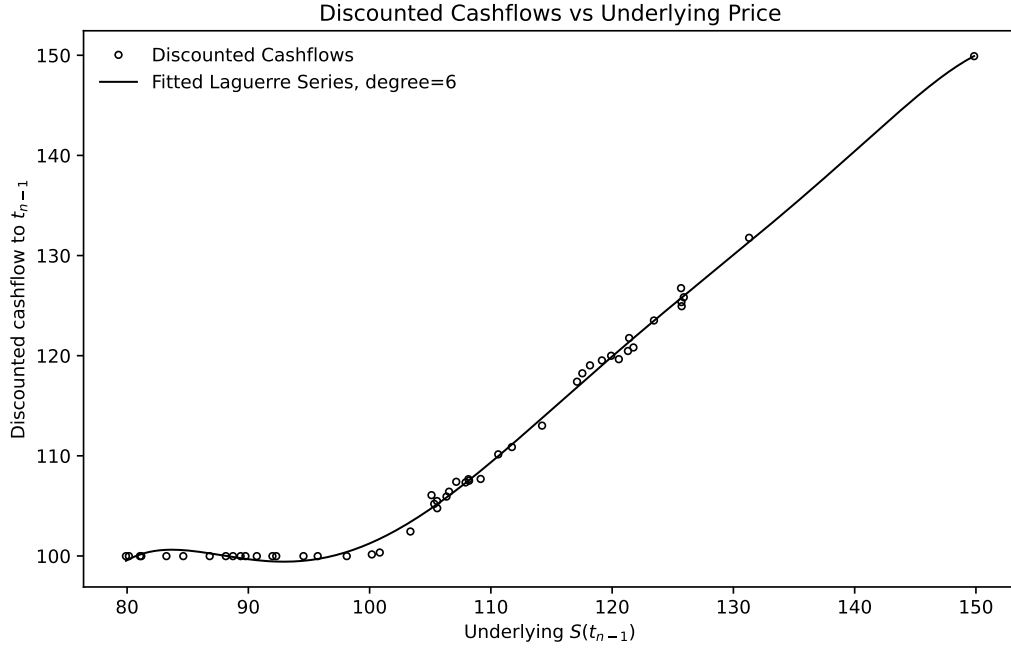


Figure 1: Discounted cashflows vs underlying price and Laguerre fit. Discounted realised cashflows at the penultimate time stop t_{n-1} plotted against $S_{t_{n-1}}$, with the fitted Laguerre-series approximation.

Figure 1 shows the discounted realised cashflow at t_{n-1} against the underlying $S_{t_{n-1}}$, with the fitted Laguerre series. We can note that when $S_{t_{n-1}} \lesssim \mathcal{P}$, the discounted cashflow is approximately flat. Economically, the redemption value F dominates because converting yields $NS \leq F$. When $S_{t_{n-1}} \gtrsim \mathcal{P}$, the cashflows increase with S and become increasingly convex. This convexity is the equity-option component embedded in the convertible payoff.

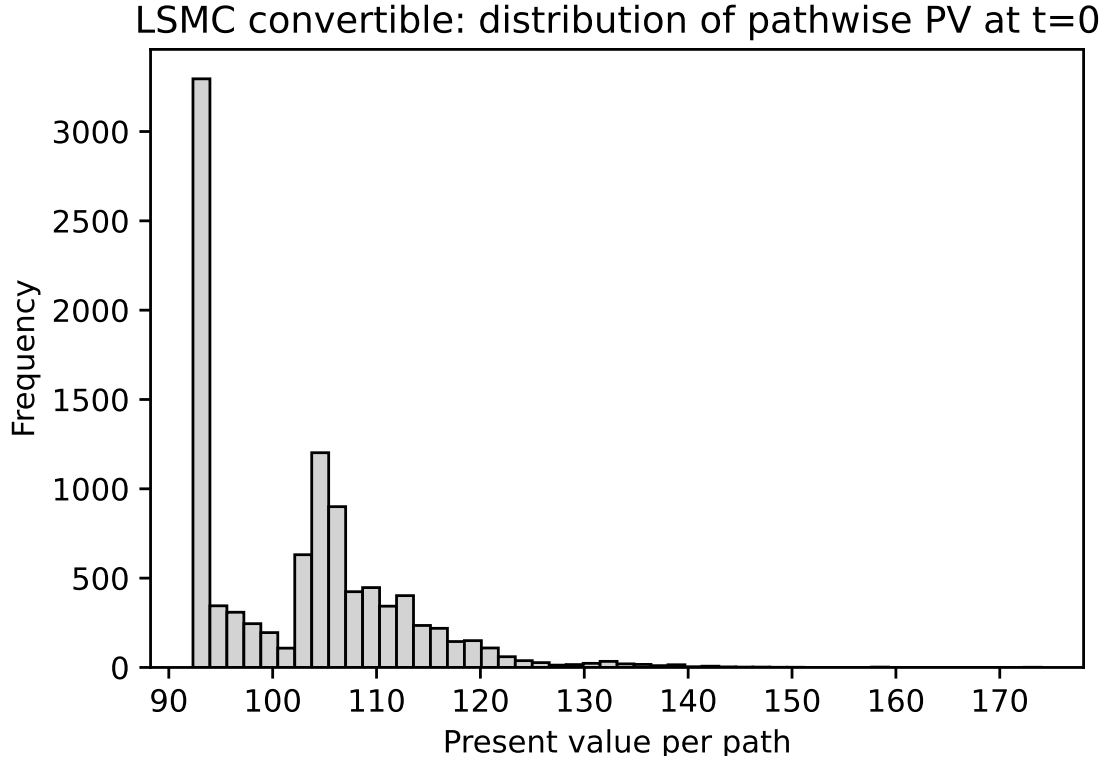


Figure 2: Distribution of pathwise present values at inception. Histogram of pathwise time-0 present values produced by the LSM algorithm. The concentration near the discounted bond floor corresponds to redemption-dominated outcomes, whereas the heavier upper tail corresponds to equity-driven outcomes where conversion becomes optimal and the payoff behaves more like NS_T

The histogram in figure 2 shows the distribution of Monte Carlo pathwise present values $V_0^{(k)}$ returned by the backward induction. A large proportion of paths cluster near a relatively low present value level. These are scenarios in which the stock does not rise sufficiently above $\mathcal{P}$, so the optimal outcome is essentially redemption at F . The time-0 value in these cases is close to the discounted bond floor Fe^{-rT} . By contrast, paths in which the stock appreciates substantially lead to conversion and hence to larger discounted payoffs that behave increasingly like NS_T . This produces a heavier right tail, i.e., as S_T rises, the payoff becomes approximately linear in S_T , and the distribution inherited the dispersion of the stock.

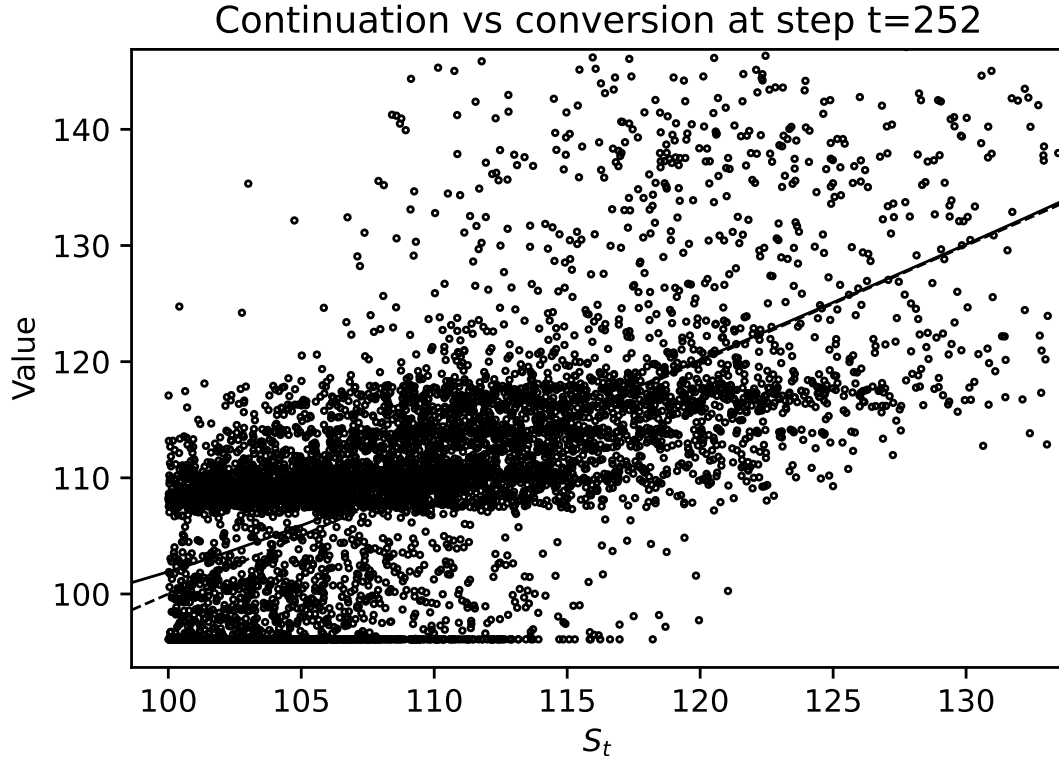


Figure 3: Continuation vs conversion at $t = 252$. Realised discounted hold values plotted against S_t , with the fitted continuation value $\hat{C}(t, S)$ and the immediate conversion value NS . Their intersection defines the estimated conversion threshold S_t^* used by the stopping rule.

We then looked at the realised discounted continuation values against S_t , together with the fitted continuation curve $\hat{C}(t, S)$ and the immediate conversion value NS , as shown in Figure 3. The large vertical spread at a given S_t reflects the fact that future stock paths can diverge substantially, so the terminal payoff depends on the current stock price, on the subsequent path, and the later stopping decisions. Hence, the continuation cashflows exhibit *substantial conditional variance* (heteroskedasticity), which typically increasing with S because higher equity levels amplify the scale of payoff dispersion.

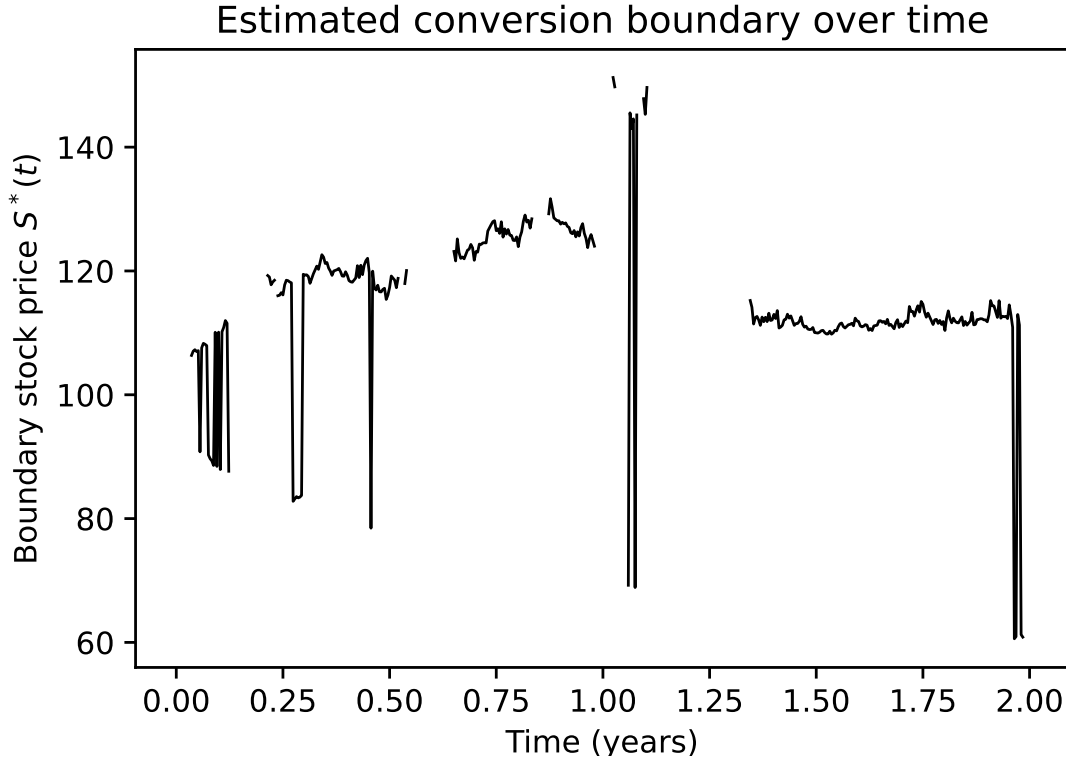


Figure 4: Estimated conversion boundary. Time series of the estimated conversion boundary S_t^* obtained by locating the stock price at which NS first exceeds the fitted continuation value $\hat{C}(t, S)$.

We then estimated the conversion boundary S_t^* extracted across time. Figure 4 shows the conversion boundary would be extremely high, with conversion effectively deferred until maturity for most scenarios, in this frictionless, noise-free idealisation for this specific zero-coupon contract. The estimated boundary is finite and shows noticeable irregularity.

The histogram shown in figure 5 shows the empirical distribution of estimated conversion times $\hat{\tau}$ across simulation paths. The mass at T indicates that the algorithm converts at maturity for most paths, following the no early exercise idea. The smaller mass of pre-maturity conversions corresponds to paths where the estimated continuation value falls below the immediate conversion value NS_{t_i} , thereby triggering early exercise.

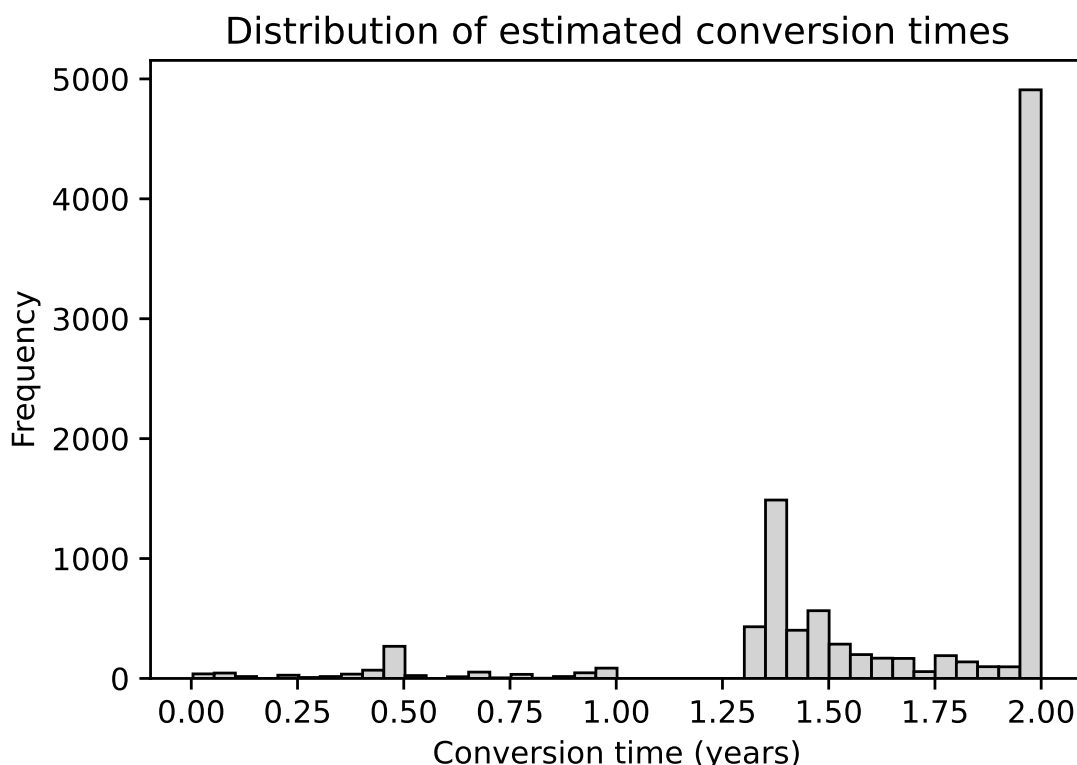


Figure 5: Distribution of estimated conversion times. Histogram of estimated conversion times $\hat{\tau}$ across Monte Carlo paths. The spike at maturity reflects that, for a zero-coupon, zero-dividend contract, optimal conversion is typically deferred until T .

5 Conclusion

In conclusion, we have demonstrated how the Longstaff-Schwartz LSMC algorithm can be applied to the pricing of convertible bonds. Monte Carlo approach showed to be a flexible tool for handling the convertible’s Bermudan-style conversion feature while capturing its hybrid debt-equity character. Figure 1 shows that the LSM regression captures the nonlinear payoff structure of the convertible, i.e., when the equity is weak, the bond floor dominates, whereas when the equity is strong, the payoff exhibits increasing convexity due to the embedded conversion option. We then saw that the histogram in 2 illustrating the two broad payoff regimes of the contract, namely debt-like outcomes concentrated near the bond floor and equity-like outcomes associated with conversion. Finally, Figure 4 shows the estimated conversion boundary over time and therefore indicates where the model infers the possibility of early conversion, as well as how stable that inferred boundary is across the exercise dates.

References

Longstaff, Francis A and Eduardo S Schwartz (2001). “Valuing American options by simulation: A simple least-squares approach”. In: *The review of financial studies* 14.1, pp. 113–147.